

Supplement for “Cressie–Read Power Divergence for Moment-Based Estimation: Hyperparameters and Finite-Sample Behavior”

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This document includes additional tables from the Monte Carlo simulations.

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Table S.1: Simulation performance metrics: second-order and first-order measures (t_{15})

DGP	n	γ	Second-order measure			First-order measure		
			Bias	MSE	Coverage distortion	Empirical SD	Mean SE	Ratio
t_{15}	25	-1.00	-0.0133	0.0636	-0.1172	0.2519	0.1872	1.3452
		-0.75	-0.0519	0.0935	-0.1550	0.3016	0.1870	1.6126
		-0.50	-0.2673	0.2755	-0.3550	0.4519	0.2025	2.2315
		-0.25	-0.4608	0.4235	-0.5020	0.4598	0.2196	2.0940
		0.00	-0.1584	0.1833	-0.2310	0.3979	0.2011	1.9784
		0.25	0.0020	0.0665	-0.1090	0.2580	0.1870	1.3797
		0.50	-0.0024	0.0714	-0.1160	0.2673	0.1863	1.4345
		0.75	-0.0060	0.0757	-0.1222	0.2753	0.1862	1.4785
		1.00	-0.0230	0.0887	-0.1392	0.2971	0.1870	1.5887
	50	-1.00	-0.0014	0.0287	-0.0470	0.1694	0.1417	1.1953
		-0.75	-0.0060	0.0315	-0.0510	0.1776	0.1424	1.2467
		-0.50	-0.1112	0.1235	-0.1500	0.3335	0.1473	2.2640
		-0.25	-0.4345	0.4244	-0.4650	0.4856	0.1643	2.9556
		0.00	-0.0396	0.0581	-0.0720	0.2379	0.1441	1.6509
		0.25	-0.0001	0.0288	-0.0460	0.1698	0.1421	1.1948
		0.50	-0.0062	0.0328	-0.0600	0.1810	0.1432	1.2643
		0.75	-0.0045	0.0382	-0.0880	0.1956	0.1435	1.3628
		1.00	-0.0161	0.0467	-0.1020	0.2156	0.1444	1.4934

Note: Results are based on 1,000 Monte Carlo replications. DGP denotes the data-generating process. n denotes the number of observations and γ the power parameter. Bias is the average of $\hat{\theta} - \theta_0$ across replications. MSE denotes the mean squared error of $\hat{\theta}$. Coverage distortion is the deviation of the empirical coverage probability from the nominal level 95%. Empirical SD is the standard deviation of $\hat{\theta}$ across replications. Mean SE is the average of the estimated standard errors. Ratio denotes the ratio of the empirical SD to the mean SE. Bias, MSE, and coverage distortion are reported as second-order performance measures because they reflect finite-sample deviations from the first-order asymptotic approximation. Empirical SD, mean SE, and their ratio summarize first-order behavior, as they evaluate the variability of the estimator and the accuracy of the first-order standard error approximation.

Table S.2: Simulation results (t_{15})

n	γ	$\hat{\theta}$	Lagrange multipliers					π				
			λ_1	λ_2	λ_3	δ	Minimum	Q1	Median	Mean	Q3	Maximum
25	-1.00	-0.0133 (0.2519)	-0.0790 (0.8187)	-0.0519 (0.3854)	0.1259 (1.1146)	-1.00E+04 (1.45E-05)	0.0244 (0.0111)	0.0336 (0.0056)	0.0373 (0.0044)	0.0400 (1.94E-09)	0.0446 (0.0039)	0.0628 (0.0369)
	-0.75	-0.0519 (0.3016)	-3.61E+10 (6.98E+11)	-3.14E+10 (5.70E+11)	2.22E+10 (3.90E+11)	-1.48E+09 (2.65E+10)	0.0227 (0.0122)	0.0323 (0.0090)	0.0359 (0.0093)	0.0379 (0.0089)	0.0425 (0.0108)	0.0564 (0.0363)
	-0.50	-0.2673 (0.4519)	-7.3E+11 (6.48E+12)	-7.4E+11 (6.38E+12)	5.24E+11 (4.49E+12)	-4.94E+10 (6.16E+11)	0.0138 (0.0139)	0.0223 (0.0164)	0.0250 (0.0182)	0.0269 (0.0188)	0.0303 (0.0218)	0.0448 (0.0561)
	-0.25	-0.4608 (0.4598)	-2.22E+12 (1.92E+13)	-2.88E+12 (3.01E+13)	2.14E+12 (2.23E+13)	-2.12E+11 (2.92E+12)	0.0081 (0.0125)	0.0143 (0.0173)	0.0159 (0.0191)	0.0172 (0.0198)	0.0189 (0.0226)	0.0319 (0.0686)
	0.00	-0.1584 (0.3979)	7.00E+01 (4.83E+02)	-1.60E+02 (8.89E+02)	5.69E+01 (3.07E+02)	-1.19E+02 (8.03E+02)	0.0182 (0.0142)	0.0289 (0.0142)	0.0319 (0.0155)	2.50E+16 (7.92E+17)	0.0366 (0.0179)	6.26E+17 (1.98E+19)
	0.25	0.0020 (0.2580)	0.0039 (0.4156)	-0.0079 (0.1195)	-0.0022 (0.1846)	-0.8371 (0.0613)	0.0213 (0.0132)	0.0352 (0.0041)	0.0390 (0.0029)	0.0400 (1.20E-09)	0.0452 (0.0035)	0.0529 (0.0148)
	0.50	-0.0024 (0.2673)	0.0076 (0.4616)	-0.0106 (0.1214)	-0.0025 (0.1832)	-0.7166 (0.0890)	0.0205 (0.0136)	0.0350 (0.0049)	0.0389 (0.0033)	0.0400 (2.92E-10)	0.0453 (0.0038)	0.0539 (0.0172)
	0.75	-0.0060 (0.2753)	0.0050 (0.4738)	-0.0066 (0.0314)	-0.0002 (0.1748)	-0.6306 (0.1143)	0.0205 (0.0138)	0.0349 (0.0054)	0.0390 (0.0032)	0.0400 (1.08E-10)	0.0453 (0.0040)	0.0535 (0.0172)
	1.00	-0.0230 (0.2971)	0.0317 (0.5282)	-0.0075 (0.0416)	-0.0014 (0.1831)	-0.5827 (0.1582)	0.0201 (0.0141)	0.0343 (0.0062)	0.0388 (0.0032)	0.0400 (8.27E-11)	0.0458 (0.0047)	0.0549 (0.0186)
50	-1.00	-0.0014 (0.1694)	0.0051 (0.3052)	1.49E-06 (0.0031)	-0.0015 (0.1142)	-1.00E+04 (2.40E-06)	0.0125 (0.0052)	0.0180 (0.0016)	0.0194 (0.0010)	0.0200 (2.84E-12)	0.0219 (0.0014)	0.0280 (0.0072)
	-0.75	-0.0060 (0.1776)	-7.72E+07 (2.44E+09)	-6.41E+07 (2.02E+09)	4.70E+07 (1.50E+09)	-2.64E+05 (8.29E+06)	0.0119 (0.0055)	0.0179 (0.0022)	0.0193 (0.0019)	0.0199 (0.0017)	0.0218 (0.0024)	0.0273 (0.0067)
	-0.50	-0.1112 (0.3335)	-1.68E+12 (3.24E+13)	-1.22E+12 (3.24E+13)	8.15E+11 (2.21E+13)	-2.93E+10 (4.08E+11)	0.0096 (0.0066)	0.0157 (0.0063)	0.0170 (0.0067)	0.0174 (0.0067)	0.0192 (0.0076)	0.0237 (0.0104)
	-0.25	-0.4345 (0.4856)	-2.92E+12 (3.34E+13)	-3.23E+12 (3.84E+13)	2.04E+12 (2.16E+13)	-3.09E+11 (4.22E+12)	0.0054 (0.0067)	0.0095 (0.0092)	0.0103 (0.0099)	0.0105 (0.0100)	0.0114 (0.0110)	0.0153 (0.0247)
	0.00	-0.0396 (0.2379)	-0.1093 (2.71E+01)	-3.4860 (5.24E+01)	2.6560 (2.19E+01)	-2.2179 (4.60E+01)	0.0108 (0.0066)	0.0177 (0.0039)	0.0190 (0.0040)	0.0192 (0.0040)	0.0209 (0.0045)	0.0246 (0.0066)
	0.25	-0.0001 (0.1698)	0.0110 (0.2677)	-0.0013 (0.0127)	-0.0052 (0.0992)	-0.8186 (0.0328)	0.0107 (0.0066)	0.0184 (0.0015)	0.0198 (0.0008)	0.0200 (7.73E-11)	0.0218 (0.0014)	0.0256 (0.0054)
	0.50	-0.0062 (0.1810)	0.0075 (0.2993)	-0.0045 (0.0217)	-0.0023 (0.0944)	-0.6929 (0.0609)	0.0103 (0.0068)	0.0183 (0.0019)	0.0197 (0.0010)	0.0200 (5.12E-11)	0.0218 (0.0015)	0.0260 (0.0074)
	0.75	-0.0045 (0.1956)	-0.0026 (0.3302)	-0.0048 (0.0267)	0.0034 (0.0920)	-0.6078 (0.0852)	0.0098 (0.0070)	0.0181 (0.0023)	0.0197 (0.0010)	0.0200 (5.93E-11)	0.0220 (0.0018)	0.0261 (0.0076)
	1.00	-0.0161 (0.2156)	0.0180 (0.3500)	-0.0058 (0.0347)	0.0016 (0.0876)	-0.5481 (0.1160)	0.0094 (0.0071)	0.0180 (0.0025)	0.0197 (0.0010)	0.0200 (4.53E-11)	0.0221 (0.0020)	0.0263 (0.0083)

Note: Results are based on 1,000 Monte Carlo replications. The data-generating process follows a t distribution with 15 degrees of freedom and mean zero. Empirical standard deviations are reported in parentheses. E denotes scientific notation. n the number of observations, and γ the power parameter. θ is the estimate of the structural parameter θ . $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers associated with the moment conditions, and δ is the Lagrange multiplier associated with the probability constraint. π denotes the estimated weights on the observations. Q1 and Q3 denote the first and third quartiles, respectively.

Table S.3: Simulation performance metrics: second-order and first-order measures

DGP	γ	Second-order metrics			First-order metrics		
		Bias	MSE	Coverage distortion	emp_sd	mean_se	se_ratio
t_5	-1.00	0.0057	0.0155	-0.0150	0.1244	0.1181	1.0530
	-0.75	-0.0010	0.0154	-0.0140	0.1240	0.1181	1.0499
	-0.50	-0.0001	0.0167	-0.0100	0.1292	0.1179	1.0959
	-0.25	-0.1110	0.1354	-0.0970	0.3510	0.1224	2.8675
	0.00	-0.0124	0.0156	-0.0030	0.1242	0.1186	1.0472
	0.25	-0.0448	0.0809	-0.0643	0.2811	0.1209	2.3245
	0.50	-0.0839	0.1367	-0.1239	0.3602	0.1228	2.9327
	0.75	-0.0877	0.1422	-0.1588	0.3669	0.1230	2.9836
	1.00	-0.0925	0.1259	-0.2038	0.3428	0.1225	2.7970
t_{15}	-1.00	0.0029	0.0123	-0.0110	0.1108	0.1041	1.0643
	-0.75	-0.0035	0.0127	-0.0190	0.1127	0.1040	1.0834
	-0.50	-0.0140	0.0225	-0.0400	0.1493	0.1040	1.4351
	-0.25	-0.2529	0.2574	-0.2480	0.4400	0.1131	3.8910
	0.00	-0.0135	0.0152	-0.0220	0.1225	0.1042	1.1757
	0.25	-0.0038	0.0128	-0.0220	0.1131	0.1040	1.0879
	0.50	-0.0081	0.0194	-0.0380	0.1390	0.1046	1.3292
	0.75	-0.0077	0.0243	-0.0640	0.1556	0.1045	1.4886
	1.00	-0.0225	0.0385	-0.0940	0.1950	0.1051	1.8547
Normal	-1.00	-0.0003	0.0093	-0.0080	0.0964	0.0963	1.0006
	-0.75	-0.0018	0.0096	-0.0140	0.0982	0.0965	1.0177
	-0.50	-0.0127	0.0201	-0.0410	0.1414	0.0970	1.4574
	-0.25	-0.3039	0.3014	-0.3380	0.4574	0.1079	4.2397
	0.00	-0.0028	0.0127	-0.0350	0.1126	0.0970	1.1608
	0.25	0.0017	0.0111	-0.0260	0.1052	0.0970	1.0843
	0.50	0.0035	0.0107	-0.0240	0.1033	0.0969	1.0668
	0.75	9.31E-05	0.0163	-0.0280	0.1277	0.0974	1.3113
	1.00	-0.0198	0.0386	-0.0680	0.1956	0.0980	1.9961

Note: Results are based on 1,000 Monte Carlo replications. DGP denotes the data-generating process. n denotes the number of observations and γ the power parameter. Bias is the average of $\hat{\theta} - \theta_0$ across replications. MSE denotes the mean squared error of $\hat{\theta}$. Coverage distortion is the deviation of the empirical coverage probability from the nominal level 95%. Empirical SD is the standard deviation of $\hat{\theta}$ across replications. Mean SE is the average of the estimated standard errors. Ratio denotes the ratio of the empirical SD to the mean SE. Bias, MSE, and coverage distortion are reported as second-order performance measures because they reflect finite-sample deviations from the first-order asymptotic approximation. Empirical SD, mean SE, and their ratio summarize first-order behavior, as they evaluate the variability of the estimator and the accuracy of the first-order standard error approximation.

Table S.4: Simulation results ($n = 100$)

DGP	γ	$\hat{\theta}$	Lagrange multipliers				π					
			λ_1	λ_2	λ_3	δ	min	q1	median	mean	q3	max
t_5	-1.00	0.0057	-0.0034	-0.0002	0.0016	-1.00E+04	0.0050	0.0095	0.0099	0.0100	0.0105	0.0166
		(0.1244)	(0.1436)	(0.0029)	(0.0404)	(1.57E-06)	(0.0028)	(0.0005)	(0.0002)	(9.05E-13)	(0.0005)	(0.0059)
		-0.75	-0.0010	-0.0013	0.0000	0.0011	-4.0025	0.0049	0.0095	0.0100	0.0100	0.0154
	-0.50	(0.1240)	(0.1281)	(0.0017)	(0.0355)	(0.0030)	(0.0029)	(0.0004)	(0.0001)	(2.92E-11)	(0.0004)	(0.0044)
		-0.0001	-1.88E+09	-1.16E+09	6.87E+08	-1.28E+07	0.0047	0.0096	0.0100	0.0100	0.0105	0.0146
		(0.1292)	(5.93E+10)	(3.65E+10)	(2.17E+10)	(4.04E+08)	(0.0030)	(0.0006)	(0.0005)	(0.0004)	(0.0006)	(0.0038)
	-0.25	-0.1110	-9.66E+11	-6.94E+11	4.30E+11	-5.49E+10	0.0042	0.0087	0.0090	0.0090	0.0094	0.0128
		(0.3510)	(1.22E+13)	(8.91E+12)	(6.0)E+12)	(7.23E+11)	(0.0032)	(0.0029)	(0.0030)	(0.0030)	(0.0031)	(0.0052)
		0.00	-0.0124	-0.0397	-0.0454	0.0453	-1.0123	0.0043	0.0096	0.0100	0.0100	0.0136
	0.25	(0.1242)	(0.9563)	(1.0132)	(1.0297)	(0.1008)	(0.0032)	(0.0005)	(0.0005)	(0.0004)	(0.0006)	(0.0026)
		-0.0448	0.0258	-0.0030	-0.0002	-0.8387	0.0041	0.0094	0.0098	0.0100	0.0105	0.0164
		(0.2811)	(0.2477)	(0.0198)	(0.0301)	(0.1441)	(0.0032)	(0.0013)	(0.0009)	(1.08E-11)	(0.0005)	(0.0173)
	0.50	-0.0839	0.0673	-0.0050	-0.0023	-0.7457	0.0040	0.0091	0.0097	0.0100	0.0106	0.0176
		(0.3602)	(0.3505)	(0.0292)	(0.0309)	(0.2601)	(0.0034)	(0.0019)	(0.0012)	(1.33E-11)	(0.0008)	(0.0189)
		0.75	-0.0877	0.0822	-0.0060	-0.0028	-0.6706	0.0038	0.0089	0.0097	0.0100	0.0167
	1.00	(0.3669)	(0.3909)	(0.0335)	(0.0320)	(0.2913)	(0.0034)	(0.0021)	(0.0011)	(3.23E-12)	(0.0011)	(0.0140)
		-0.0925	0.1070	-0.0139	-0.0030	-0.6125	0.0036	0.0088	0.0097	0.0100	0.0110	0.0162
		(0.3428)	(0.4176)	(0.0614)	(0.0335)	(0.2804)	(0.0034)	(0.0022)	(0.0009)	(2.46E-11)	(0.0014)	(0.0105)
t_{15}	-1.00	0.0029	-0.0038	-8.82E-05	0.0013	-1.00E+04	0.0064	0.0094	0.0099	0.0100	0.0106	0.0146
		(0.1108)	(0.1746)	(0.0020)	(0.0585)	(1.11E-06)	(0.0024)	(0.0005)	(0.0002)	(2.16E-12)	(0.0005)	(0.0043)
		-0.75	-0.0035	-0.0052	3.20E-05	0.0017	-4.0019	0.0062	0.0094	0.0099	0.0100	0.0142
	-0.50	(0.1127)	(0.1708)	(0.0015)	(0.0572)	(0.0026)	(0.0026)	(0.0005)	(0.0002)	(5.21E-11)	(0.0005)	(0.0037)
		-0.0140	-1.40E+09	-1.16E+09	7.94E+08	-6.89E+07	0.0058	0.0093	0.0098	0.0099	0.0105	0.0138
		(0.1493)	(2.86E+10)	(2.37E+10)	(1.62E+10)	(1.48E+09)	(0.0028)	(0.0011)	(0.0011)	(0.0011)	(0.0013)	(0.0037)
	-0.25	-0.2529	-1.02E+12	-9.7543E+11	5.70E+11	-9.26E+10	0.0042	0.0071	0.0074	0.0075	0.0079	0.0105
		(0.4400)	(1.61E+13)	(1.64E+13)	(8.31E+12)	(2.21E+12)	(0.0034)	(0.0041)	(0.0043)	(0.0043)	(0.0046)	(0.0067)
		0.00	-0.0135	-0.0797	-0.1202	0.1710	-1.0165	0.0055	0.0094	0.0099	0.0100	0.0132
	0.25	(0.1225)	(1.5153)	(2.2368)	(3.3629)	(0.1529)	(0.0030)	(0.0007)	(0.0006)	(0.0006)	(0.0008)	(0.0028)
		-0.0038	0.0005	-0.0006	0.0006	-0.8092	0.0053	0.0095	0.0100	0.0100	0.0106	0.0131
		(0.1131)	(0.1645)	(0.0051)	(0.0518)	(0.0171)	(0.0031)	(0.0005)	(0.0002)	(1.17E-11)	(0.0005)	(0.0026)
	0.50	-0.0081	0.0051	-0.0022	0.0003	-0.6820	0.0052	0.0094	0.0099	0.0100	0.0106	0.0131
		(0.1390)	(0.2021)	(0.0132)	(0.0506)	(0.0504)	(0.0033)	(0.0008)	(0.0004)	(2.70E-11)	(0.0006)	(0.0034)
		0.75	-0.0077	0.0078	-0.0035	0.0002	-0.5938	0.0051	0.0094	0.0099	0.0100	0.0131
	1.00	(0.1556)	(0.2342)	(0.0221)	(0.0487)	(0.0924)	(0.0034)	(0.0010)	(0.0004)	(1.37E-11)	(0.0007)	(0.0044)
		-0.0225	0.0416	-0.0115	-0.0004	-0.5435	0.0047	0.0092	0.0099	0.0100	0.0108	0.0137
		(0.1950)	(0.3186)	(0.0727)	(0.0493)	(0.1694)	(0.0034)	(0.0013)	(0.0005)	(7.00E-11)	(0.0009)	(0.0070)
Normal	-1.00	-0.0003	-0.0051	-0.0003	0.0019	-1.00E+04	0.0070	0.0094	0.0099	0.0100	0.0106	0.0139
		(0.0964)	(0.1889)	(0.0027)	(0.0705)	(9.51E-07)	(0.0021)	(0.0005)	(0.0002)	(3.07E-12)	(0.0005)	(0.0039)
		-0.75	-0.0018	-0.0010	-3.09E-05	0.0003	-4.0016	0.0069	0.0094	0.0099	0.0100	0.0136
	-0.50	(0.0982)	(0.1862)	(0.0018)	(0.0696)	(0.0022)	(0.0022)	(0.0005)	(0.0002)	(6.16E-11)	(0.0005)	(0.0034)
		-0.0127	-2.77E+12	-2.53E+12	1.95E+12	-6.27E+10	0.0065	0.0092	0.0097	0.0098	0.0105	0.0132
		(0.1414)	(8.32E+13)	(7.68E+13)	(5.94E+13)	(1.92E+12)	(0.0025)	(0.0013)	(0.0013)	(0.0013)	(0.0014)	(0.0034)
	-0.25	-0.3039	-7.5E+11	-6.92E+11	4.93E+11	1.59E+11	0.0041	0.0062	0.0066	0.0067	0.0071	0.0092
		(0.4574)	(6.59E+12)	(6.57E+12)	(4.49E+12)	(4.44E+12)	(0.0036)	(0.0044)	(0.0047)	(0.0047)	(0.0051)	(0.0068)
		0.00	-0.0028	-0.0714	-0.1103	0.1817	-1.0118	0.0062	0.0094	0.0099	0.0100	0.0131
	0.25	(0.1126)	(1.9666)	(2.9020)	(4.9557)	(0.1182)	(0.0027)	(0.0006)	(0.0005)	(0.0004)	(0.0007)	(0.0025)
		-0.0017	-0.0024	-0.0003	0.0009	-0.8084	0.0059	0.0094	0.0100	0.0100	0.0107	0.0130
		(0.1052)	(0.1847)	(0.0035)	(0.0702)	(0.0111)	(0.0028)	(0.0005)	(0.0001)	(2.00E-11)	(0.0005)	(0.0022)
	0.50	0.0035	-0.0034	-0.0008	0.0010	-0.6766	0.0058	0.0094	0.0100	0.0100	0.0107	0.0127
		(0.1033)	(0.1835)	(0.0067)	(0.0664)	(0.0149)	(0.0030)	(0.0005)	(0.0002)	(2.42E-11)	(0.0005)	(0.0020)
		0.75	0.0001	0.0062	-0.0035	-0.0004	-0.5891	0.0057	0.0094	0.0099	0.0100	0.0129
	1.00	(0.1277)	(0.2262)	(0.0348)	(0.0614)	(0.0770)	(0.0032)	(0.0008)	(0.0003)	(1.55E-11)	(0.0007)	(0.0046)
		-0.0198	0.0447	-0.0191	0.0025	-0.5477	0.0055	0.0092	0.0099	0.0100	0.0108	0.0138
		(0.1956)	(0.3627)	(0.1026)	(0.0595)	(0.2061)	(0.0034)	(0.0014)	(0.0006)	(7.73E-11)	(0.0009)	(0.0089)

Note: Results are based on 1,000 Monte Carlo replications. The data-generating process follows either a Student's t distribution with 5 or 15 degrees of freedom and mean zero, or a normal distribution with mean zero and unit variance. Empirical standard deviations are reported in parentheses. E denotes scientific notation. n the number of observations, and γ the power parameter. $\hat{\theta}$ is the estimate of the structural parameter θ . $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers associated with the moment conditions, and δ is the Lagrange multiplier associated with the probability constraint. π denotes the estimated weights on the observations. Q1 and Q3 denote the first and third quartiles, respectively.